Fiscal Year:	FY 2016	**Task Last Updated:**	FY 05/02/2016
PI Name:	Stanton, Patric Ph.D.		
Project Title:	Effects of Space Radiation on Long-Term Synaptic Plasticity and Neurogenesis in Normal and Alzheimer's Disease Transgenic Mice		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	10595-1690	**Congressional District:**	18
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Space Radiobiology NNJ12ZSA001N
Start Date:	11/01/2012	**End Date:**	10/31/2017
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: Extended to 10/31/2017 per S. Monk/LaRC (Ed., 9/20/16) NOTE: Extended to 10/31/2016 per S. Monk/LaRC (Ed., 9/14/15)		
Key Personnel Changes/Previous PI:	October 2015 report: No Changes.		
COI Name (Institution):	Blakely, Eleanor (Lawrence Berkeley National Laboratory)		
Grant/Contract No.:	NNX13AB66G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The hippocampus and dentate gyrus are critically important brain regions necessary for the formation of long-term memories, and damage to these regions is a cause of memory deficits in early Alzheimer's disease patients. Both regions are also notable as sites where stem cells can differentiate into new neurons throughout life, a process called 'neurogenesis.' An abundance of data indicate that exposure to even low doses of ionizing radiation prevalent in space travel (SR) results in impairments in learning and long-term reduction in hippocampal neurogenesis. It is unknown how radiation causes these impairments, whether they predispose individuals to or accelerate the development of Alzheimer's disease, and how we can prevent such sequelae. The key, central objectives of this proposal are to 1) examine the long-term impact of SR on electrical communication between neurons, and changes in function and fine structure of the sites known as 'dendritic spines' where neurons form synaptic contacts to communicate with one another; 2) determine whether SR, in reducing neurogenesis, also alters the functionality of newly-born neurons; and 3) determine whether, in the most widely-studied animal model of human Alzheimer's disease, SR differentially affects electrical or physical function of neurons, and/or accelerates the disease process. The methods we propose to accomplish these goals are 1) electrical recordings from living slices of mouse brain, and optical imaging of minute compartments in neurons known as dendritic spines; 2) in vivo injection of fluorescent markers of newly-born neurons that arise from the continual process of neurogenesis; and 3) comparison by the same methods of the effects of SR in slices from normal wildtype mice to slices from the most widely-used transgenic mouse model of human Alzheimer's disease. Our proposed studies directly address multiple key objectives in this RFP (request for proposals), namely to determine dependencies of late central nervous system (CNS) risks such as early onset of dementia or Alzheimer's disease, determine if SR causes early molecular, cellular, and tissue environment changes in hippocampus indicative of dementia, and to develop new biological models of Alzheimer's disease or other late forms of dementia that occur in humans. The existing knowledge gap is immense and presents a major obstacle to rational assessment of short- and long-term risk to the central nervous system of low-dose radiation exposure in space. Our experiments will, for the first time, examine the mechanisms by which SR impairs synaptic function in normal brain, assess whether SR does, in fact, enhance long-term risk of Alzheimer's disease, and fuel the search for countermeasures to reduce this risk, towards making the environment of space safe for long-term exploration.		
Rationale for HRP Directed Research:			
Research Impact/Earth Benefits:	This proposal addresses key questions of susceptibility of normal and diseased brain to subtle yet critical long-term damage from radiation. For humans to spend long periods in space, we must clearly understand the CNS risks of space radiation to brain function, using direct measures of synaptic function, plasticity underlying cognitive learning and neuronal replacement, and identify effective countermeasures. However, the brain is very susceptible to radiation, including medical and occupational exposure on Earth. There are pluripotent stem cells in the CNS, and key brain regions experience replacement of neurons by stem cells throughout life. Since stem cells are very sensitive to radiation, even low dose radiation could cause subtle alterations in CNS function, and simultaneously impair its ability to repair damage. If radiation accelerates the development of neurodegenerative diseases such as Alzheimers disease and alters brain cognitive performance during a mission, effective countermeasures will be critical to mission success. Transgenic mouse models of Alzheimers disease develop the disease in months (far more rapidly than humans), enabling us to determine whether radiation exposure accelerates disease progression, and begin testing antioxidants as potential protectants.		

Task Progress:	Aim 1 activities: The objective of this aim is to examine the long-term impact of space radiation on electrical communication between neurons, and changes in function and fine structure of the sites known as 'dendritic spines,' the sites where neurons form synapses to communicate with one another. In Year 03, hippocampal slice studies from normal C57Bl/6 mice irradiated with 56Fe and 28Si sources in 2013 were used to assess the 6-7 month and 20 month post-irradiation impact of exposure on activity-dependent long-term synaptic plasticity and hippocampal-dependent cognitive learning in Barnes maze spatial learning and memory retention, and active-avoidance spatial memory tests. Long-term potentiation (LTP) and learning acquisition exhibited interesting sex-specific and ion-specific differences at 6-7 months post-exposure. 28Si markedly reduced the magnitude of LTP in both male and female mice, but it did not alter biochemical mechanisms of synaptic plasticity that are downstream of synaptic activation, indicating that the effects of 28Si are focused on synaptic transmission necessary for NMDA receptor-dependent changes in synaptic strength. Moreover, while these impairments in LTP were correlated with impaired learning in male mice, learning rate was not altered in female mice, suggesting the presence of sex-specific compensatory mechanisms that can support learning in females. Finally, we also found that 28Si, at 20 months post-exposure, resulting in the same enhancement in learning that 56Fe did in our previous year experiments, suggests the existence of long-term compensatory mechanisms in both male and female mice that can respond to low-dose high-energy particle radiation exposure (LDRp) exposure given enough time. Aim 2 activities: The objective of this aim is to determine whether neurogenesis, that is, ongoing birth and maturation of new neurons, is persistently affected in the hippocampal dentate gyrus by space radiation exposure. The dentate gyrus was selected because it is a brain region where neurogenesis continues throughout life, demonstrating intriguing potential for CNS repair that is poorly understood. We have begun exposure to LDRp of a transgenic mouse strain that expresses YFP under control of the tamoxifen promoter (Cheng et al., 2011). These mice (DCX-CreERT2 F18 and F30) contain a tamoxifen-inducible recombinase driven by a BAC-based promoter of doublecortin, a marker for newborn neurons. These mice will be used to assess the time course of impairments in neurogenesis and, more importantly, the functional membrane properties and synaptic innervation of newly-born neurons and its sensitivity to space radiation. Once enough mice are available, we will also attempt to cross these DCX-CreERT2 mice with Tg2576 mice, to produce an animal that will develop Alzheimer's disease (AD) and have newly-born neurons that can be fluorescently labeled in a pulse-chase manner to assess the short and long-term effects of radiation exposure on neurogenesis in animals which are going to develop AD. These will be extraordinary new tools for the study of the effects of LDRp on neurogenesis and the importance of these effects to AD progression. In the mean time, the surprising data we collected showing enhancement of LTP and better performance in memory tasks 20 months following exposure to LDRp led us to question whether the abolishment of neurogenesis reported following LDRp persists chronically. In the previous grant year, we demonstrated the feasibility of these methods for the proposed studies. We are now quantifying doublecortin (DCX)-positive neuroblasts and newborn neurons in the subgranular zone and granule layer of the dentate gyrus (both ventral and dorsal hippocampus) in LDRp-exposed mice and age matched controls 6 and 20 months post exposure. Aim 3 activities: The objective of this aim is to determine whether, in the Tg2576 mouse, the most widely-studied animal model of human Alzheimer's disease, SR differentially affects electrical or physical function of neurons, and/or accelerates the disease process. In Year 01 we exposed a cohort of Tg2576 mice to a 56Fe source. These animals were used in year 02 to perform studies on basal synaptic transmission and activity-dependent long-term synaptic plasticity. Chronic effects of LDRp, 20 months post-exposure, on synaptic plasticity were assessed following chemically-induced (Forskolin + Rolipram, 10 ÂµM each) cyclic AMP-dependent, or theta-burst stimulus-induced long-term potentiation (LTP) of Schaffer collateral-CA1 synapses in hippocampal slices. LDRp produced a marked enhancement in the magnitude of stimulus-evoked LTP and chemically-evoked cAMP-dependent LTP ($P < 0.05$, repeated-measures ANOVA) in wildtype mice. However, this enhancement was not observed in Tg2576 mice, consistent with an interaction between AD and the effects of LDRp on synaptic plasticity. Reference: Cheng X, Li Y, Huang Y, Feng X, Feng G, Xiong ZQ (2011) Puls labeling and long-term tracing of newborn neurons in the adult subgranular zone. Cell Res., 21:338-349.
Bibliography Type:	Description: (Last Updated: 03/10/2021)
Abstracts for Journals and Proceedings	Miry O, Gopaul KR, Zhang XL, Tagge CA, Moncaster JA, Goldstein LE, Stanton PK. "Long-Term Effects of Galactic Cosmic Radiation on Adult Neurogenesis, Hippocampal Plasticity, Learning and Memory." Presented at the 2015 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 13-15, 2015. 2015 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 13-15, 2015. , Jan-2015
Abstracts for Journals and Proceedings	Miry O, Subah G, Gopaul K, Vose LR, Zhang XL, Moncaster J, Goldstein LE, Stanton PK. "Long-Term Effects of Galactic Cosmic Radiation on Synaptic Plasticity, Learning and Memory." Presented at the 45th Society for Neuroscience Congress, Chicago, IL, October 17-21, 2015. Neuroscience 2015, Chicago, IL, October 17-21, 2015. Program#/Poster#:445.05/CC20. Available at: http://www.abstractsonline.com/Plan/ViewAbstract.aspx?sKey=866e37bd-32e8-4d71-ab0a-2d97dd8096af&cKey=b1696b0d-cfa9-4815-8063-6df575a52d7b&mKey=d0ff4555-8574-4fbb-b9d4-04eec8ba0c84 ; accessed 5/3/2016. Society for Neuroscience Abstracts. 2015 Oct;455.05. , Oct-2015
Abstracts for Journals and Proceedings	Miry O, Subah G, Gopaul KR, Vose LR, Zhang XL, Moncaster JA, Goldstein LE, Stanton PK. "Effects of Space Radiation on Long-Term Synaptic Plasticity and Neurogenesis in Normal and Alzheimers Disease Transgenic Mice." Presented at the 2016 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 8-11, 2016. 2016 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 8-11, 2016. , Feb-2016
Abstracts for Journals and Proceedings	Vose LR, Miry O, Zhang XL, Stanton PK. "Effect of thyroxine on synaptic function and neurogenesis following galactic cosmic radiation." Presented at the 2016 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 8-11, 2016. 2016 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 8-11, 2016. , Feb-2016